



**KNOX GRAMMAR SCHOOL**  
**MATHEMATICS FACULTY**

**Set By: IM**

**Teachers:**

DS  
IM  
IB (Year 12)  
IB (Year 11)  
RM (Year 11)  
SH  
LS

**2005**

**TRIAL HSC EXAMINATION**

# Mathematics (Year 12)

## General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using blue or black pen
- Board-approved calculators may be used
- A table of standard integrals is provided
- All necessary working should be shown in every question

## Total marks (120)

- Attempt Questions 1–10
- All questions are of equal value
- Use a **SEPARATE** Writing Booklet for each question
- Write your **Board of Studies Student Number and Class Teacher's Initials** on the front cover of each of your writing booklets

**Board of Studies Student Number:** \_\_\_\_\_

**Class Teacher's Initials:** \_\_\_\_\_

(Year 12)  
(Year 11)  
11

(Year 12)  
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(Year 11)  
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**Total marks (120)**

**Attempt questions 1 – 10**

**All questions are of equal value**

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

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**Question 1 (12 marks)**

**Marks**

- (a) Evaluate  $\frac{3}{\log_e 9}$  correct to three significant figures. **2**
- (b) Differentiate with respect to  $x$  the expression  $\frac{1}{2x} + \pi$ . **2**
- (c) Simplify  $\frac{2x}{3} - \frac{x+2}{5}$ . **2**
- (d) Find a primitive of  $2 - \sqrt{x}$ . **2**
- (e) Solve  $1 - 9x^2 = 0$ . **2**
- (f) Peter paid \$1.15 per litre of petrol. This was 12.5% above the recommended retail price. Calculate the recommended retail price per litre of petrol. **2**

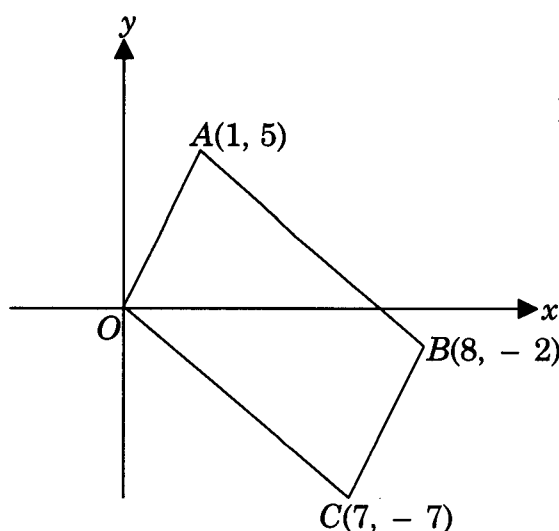


Diagram is not to scale

In the diagram,  $OABC$  is a quadrilateral where the coordinates of  $O$ ,  $A$ ,  $B$  and  $C$  are  $(0, 0)$ ,  $(1, 5)$ ,  $(8, -2)$  and  $(7, -7)$  respectively.

- |     |                                                                                             |   |
|-----|---------------------------------------------------------------------------------------------|---|
| (a) | Find the midpoint of the interval joining $AC$ .                                            | 1 |
| (b) | Find the slope of $AB$ .                                                                    | 1 |
| (c) | Show that the equation of the line containing $AB$ is $x + y = 6$ .                         | 1 |
| (d) | Find the exact length of $AB$ .                                                             | 2 |
| (e) | Show that $OC$ is parallel to $AB$ .                                                        | 1 |
| (f) | Explain why $OABC$ is a parallelogram.                                                      | 2 |
| (g) | Find the perpendicular distance from $O$ to $AB$ . Leave your answer in simplest surd form. | 2 |
| (h) | Find the area of parallelogram $OABC$ .                                                     | 2 |

**Question 3 (12 marks)** Use a SEPARATE writing booklet

**Marks**

(a) Find:

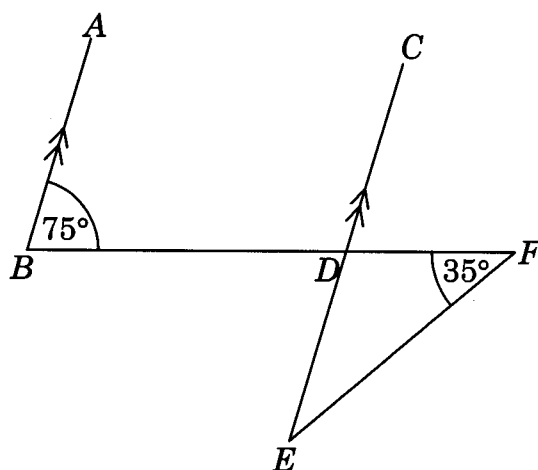
(i)  $\frac{d}{dx}(x^2 e^{3x})$ .

**2**

(ii)  $\frac{d}{dx}\left(\frac{\sin x}{x}\right)$ .

**2**

(b)



**Diagram is not to scale**

In the diagram,  $AB \parallel CE$ ,  $\angle ABF = 75^\circ$  and  $\angle BFE = 35^\circ$ .

**2**

Find the size of  $\angle DEF$ , giving reasons.

(c) The graph of  $y = f(x)$  passes through the point  $(2, 12)$  and  $f'(x) = 9x^2 + 4$ .

**2**

Find  $f(x)$ .

(d) Find  $\int \frac{4x}{x^2 - 3} dx$ .

**2**

(e) Find  $\int_{\frac{\pi}{6}}^{\pi} \cos 2x \, dx$ .

**2**

- (a) The equation of a parabola is  $(x+3)^2 = -12(y-1)$ .
- (i) Write down the coordinates of the vertex of this parabola. 1
- (ii) State the coordinates of the focus. 1
- (iii) State the equation of the directrix. 1
- (iv) Sketch the graph of the parabola, showing the above features. 2
- (b) (i) Express  $0.\dot{2}\dot{5}$  as an infinite geometric series. 1
- (ii) Hence, write  $0.\dot{2}\dot{5}$  as a rational number in lowest terms. 1

(c)

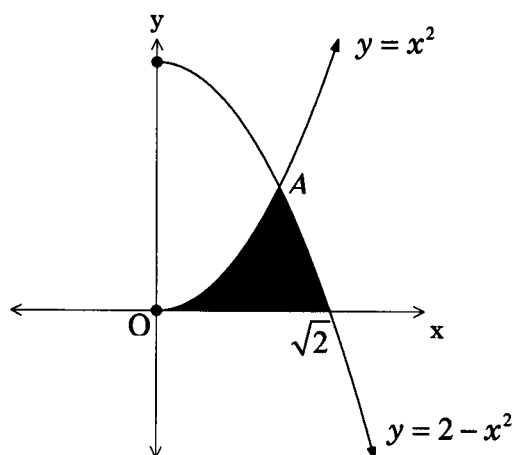


Diagram is not to scale

The diagram above shows the graphs of  $y = x^2$  and  $y = 2 - x^2$  intersecting at the point A. The curve  $y = 2 - x^2$  crosses the x-axis at  $(\sqrt{2}, 0)$ .

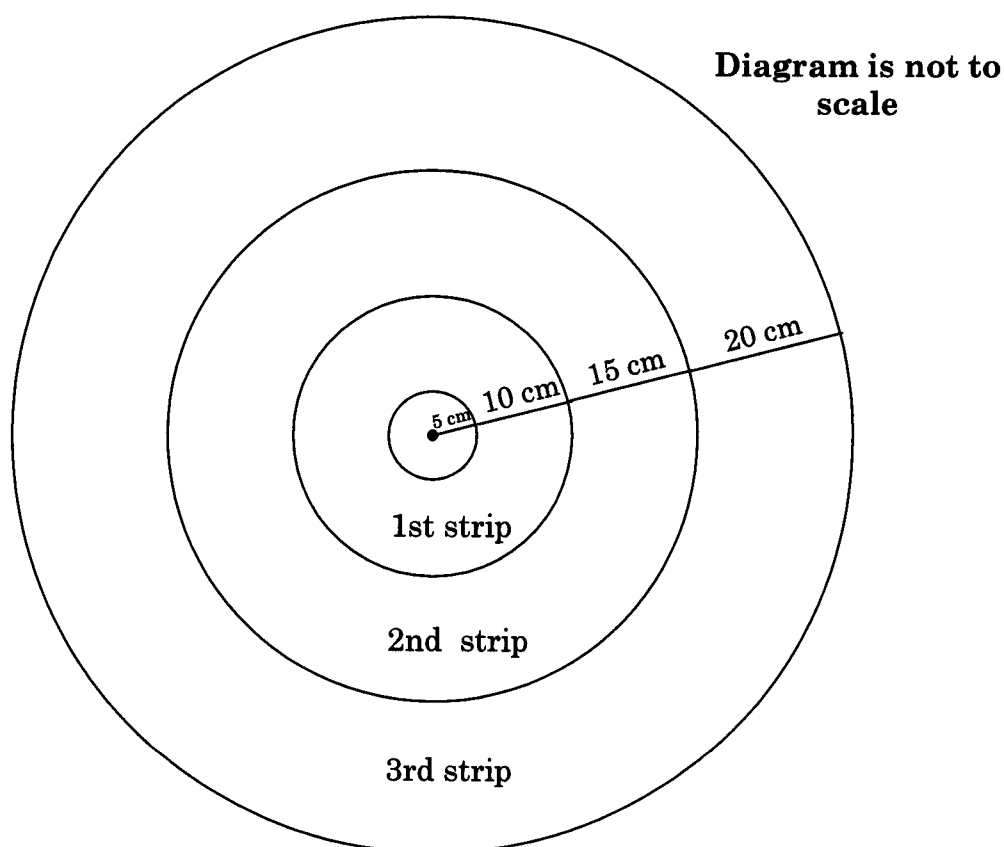
Note that  $x \geq 0$ .

- (i) Find the x-coordinate of A. 1
- (ii) Find the area of the shaded region bounded by  $y = x^2$ ,  $y = 2 - x^2$  and the x-axis. Give your answer correct to two decimal places. 4

(a) Consider the function  $f(x) = 2x^3 - 6x^2 - 18x + 1$ .

- |       |                                                                                                   |   |
|-------|---------------------------------------------------------------------------------------------------|---|
| (i)   | Find the coordinates of the stationary points on the curve $y = f(x)$ and determine their nature. | 4 |
| (ii)  | Find the values of $x$ for which $y = f(x)$ is increasing.                                        | 2 |
| (iii) | For what value of $x$ is the rate of decrease the greatest.                                       | 1 |

(b)



Beginning with a circular piece of fabric of radius 5 cm, Susan sewed together circular strips of different coloured fabrics which increased in width to make a circular table cloth as shown in the diagram. The finished width of the first strip was 10 cm and of the second strip was 15 cm and so on.

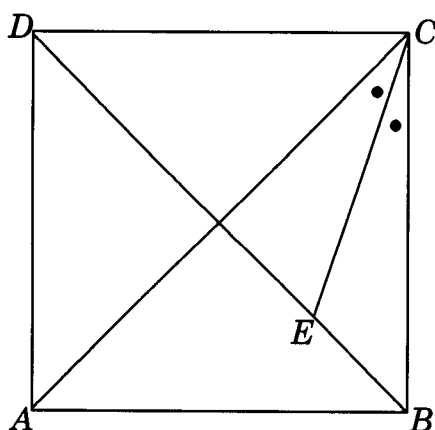
- |      |                                                                                            |   |
|------|--------------------------------------------------------------------------------------------|---|
| (i)  | Show that the width of the 10 <sup>th</sup> strip was 55 cm.                               | 2 |
| (ii) | How many strips must be sewn together to complete a circular table cloth of radius 455 cm? | 3 |

(a) Consider the equation  $\log_e y = x \log_e \left(\frac{1}{2}\right)$ , where  $y > 0$ .

(i) Write down an expression for  $y = f(x)$ . 2

(ii) Sketch the graph of  $y = f(x)$  and label the axes appropriately. 1

(b)



**Diagram is not to scale**

In the diagram,  $ABCD$  is a square with diagonals  $AC$  and  $BD$ . The point  $E$  lies on  $DB$  and the interval  $CE$  bisects  $\angle ACB$ .

**Copy or trace this diagram into your writing booklet.**

(i) Prove, with reasons, that  $\angle DCE = \angle DEC$ . 2

(ii) Hence, show that  $DE = DA$ . 2

(c) An object falls into a tank filled with oil. The rate of decrease of the velocity is proportional to its velocity  $v$  centimetres per second. This statement can be expressed mathematically by the equation  $\frac{dv}{dt} = -kv$ , where  $k$  is a constant and  $t$  is time measured in seconds.

The initial speed of the object when it enters the tank is 85 cm/s. Five seconds later, the speed of the object is 60 cm/s.

(i) Verify that  $v = Ae^{-kt}$ , where  $A$  is a constant, satisfies  $\frac{dv}{dt} = -kv$ . 1

(ii) Find the values of  $A$  and  $k$ . 3

(iii) Calculate the velocity of the object when  $t = 8$ . 1

(a)

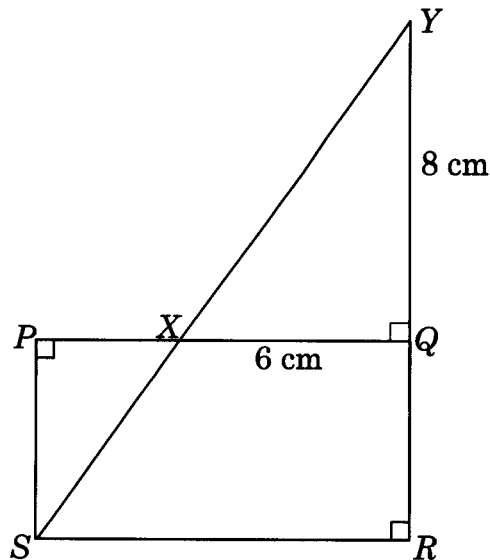


Diagram is not to scale

In the diagram,  $PQRS$  is a rectangle and  $SR = 3PS$ . The points  $R$ ,  $Q$  and  $Y$  are collinear points.  $XQ = 6$  cm and  $YQ = 8$  cm.

(i) Prove that  $\triangle PXS \parallel \triangle QXY$ .

2

(ii) Find the length of  $PS$ .

2

(b) The velocity of a particle moving in a straight line is given by  $v = 1 - 2\cos t$  for  $0 \leq t \leq 2\pi$ , where  $v$  is measured in metres per second and time  $t$  is measured in seconds.

(i) At what times in the interval  $0 \leq t \leq 2\pi$ , is the particle at rest?

2

(ii) Sketch the graph of  $v$  against  $t$  for  $0 \leq t \leq 2\pi$ .

2

(iii) What is the maximum velocity of the particle in the interval  $0 \leq t \leq 2\pi$ ?

1

(iv) Calculate the total distance travelled by the particle in the first  $\pi$  seconds.

3

**Question 8 (12 marks)** Use a SEPARATE writing booklet

**Marks**

(a) Consider the function  $y = \ln(x - 2)$  for  $x > 2$ .

(i) Sketch the graph of  $y = \ln(x - 2)$ , showing essential features.

**2**

(ii) Use Simpson's rule with three function values to evaluate:

**2**

$$\int_3^5 \ln(x - 2) \, dx$$

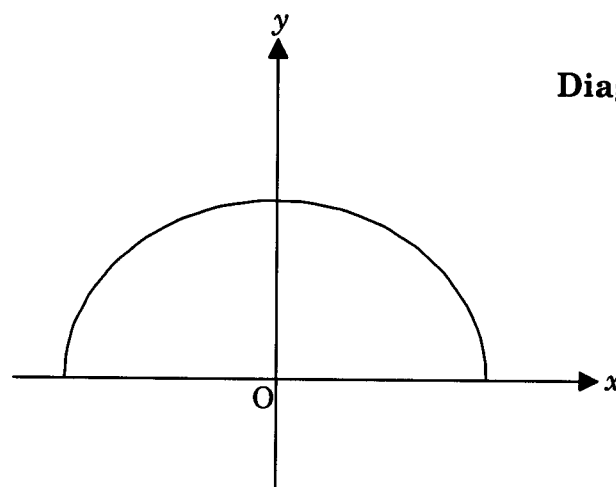
Round off your answer to one decimal place.

(b) The quadratic equation  $x^2 + mx + n = 0$  has one root that is twice the other. Let one of the roots be  $\alpha$ .

**4**

Using the equations for the sum and product of the roots of  $x^2 + mx + n = 0$ , find the value of  $\frac{m^2}{n}$ .

(c)



The diagram shows part of the curve with equation  $\frac{x^2}{9} + \frac{y^2}{4} = 1$  which lies above the  $x$ -axis.

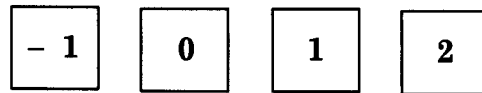
(i) Find the  $x$ -intercepts.

**1**

(ii) Find the volume of the solid of revolution formed when this curve is rotated about the  $x$ -axis. Leave your answer in simplified form in terms of  $\pi$ .

**3**

- (a) Two cards are chosen at random from the four cards shown below.



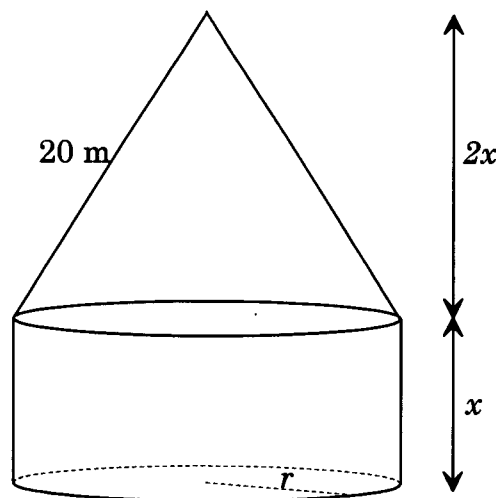
A person holds the two chosen cards in their hand.

- |      |                                                                                    |   |
|------|------------------------------------------------------------------------------------|---|
| (i)  | Calculate the probability that the sum of the numbers on the cards chosen is zero. | 1 |
| (ii) | Find the probability that at least one of the cards in the person's hand is a 2.   | 1 |
- (b) A water tank which is initially full is being emptied. The remaining quantity of water,  $L$  litres, in the tank, at any time,  $t$  minutes, after it starts to empty is given by  $L = 1000(20 - t)^2$  for  $0 \leq t \leq 20$ .
- |       |                                                                                          |   |
|-------|------------------------------------------------------------------------------------------|---|
| (i)   | How much water was in the tank initially?                                                | 1 |
| (ii)  | Show that $\frac{dL}{dt} = 2000t - 40\,000$ .                                            | 1 |
| (iii) | At what time is the water flowing out of the tank at a rate of 20 000 litres per minute? | 1 |
| (iv)  | How long will it take to empty the tank?                                                 | 1 |
| (iv)  | At what rate is the tank emptying when the tank is half full?                            | 2 |
- (c) Consider the function  $f(x) = \pi x - \cos \pi x$ .
- |      |                                                                                                                    |   |
|------|--------------------------------------------------------------------------------------------------------------------|---|
| (i)  | Find $f''\left(\frac{1}{2}\right)$ .                                                                               | 2 |
| (ii) | Show that the point $\left(\frac{1}{2}, \frac{\pi}{2}\right)$ is a point of inflexion on the graph of $y = f(x)$ . | 2 |

**Question 10 (12 marks)** Use a SEPARATE writing booklet**Marks**

- (a) An initial sum of \$2000 is invested into a retirement fund and interest is compounded at the rate of 8% per annum, every six months.
- (i) If no further deposits are made into this fund, calculate how much the initial investment is worth at the end of 20 years? **2**
- (ii) Now suppose that at the beginning of the second year, and the beginning of each subsequent year, a further \$500 is deposited into the fund. Determine how much the fund is worth at the end of 20 years? **4**

(b)



A grain silo has a cylindrical shaped wall and a cone shaped roof as shown in the diagram. Let the radius of the base of the silo be  $r$  metres and the height of the cylinder be  $x$  metres. The height of the cone is  $2x$  metres.

- (i) Show that if the length of the slant edge of the cone is 20 metres, then  $r^2 = 400 - 4x^2$ . **1**
- (ii) Show that the volume,  $V$ , in cubic metres, of the silo is given by: **2**

$$V = \frac{20\pi}{3}(100x - x^3)$$

- (iii) Find the exact height of the silo so that it holds a maximum amount of grain. **3**

**End of Paper**

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## STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln(x + \sqrt{x^2 - a^2}), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln(x + \sqrt{x^2 + a^2})$$

Note  $\ln x = \log_e x, \quad x > 0$

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QUESTION 1 (DS)

a)  $\frac{3}{\log_e 4} = 1.3653588 \dots \textcircled{1}$

$= 1.37 \text{ (3 S.F.) } \textcircled{1}$

b)  $\frac{d}{dx} \left( \frac{1}{2x} + \pi \right) = \frac{d}{dx} \left( \frac{1}{2} x^{-1} + \pi \right)$   
 $= -\frac{1}{2} x^{-2}$   
 $= -\frac{1}{2x^2} \textcircled{2}$

c)  $\frac{2x}{3} - \frac{x+2}{5} = \frac{10x - 3(x+2)}{15}$   
 $= \frac{7x-6}{15} \textcircled{2}$

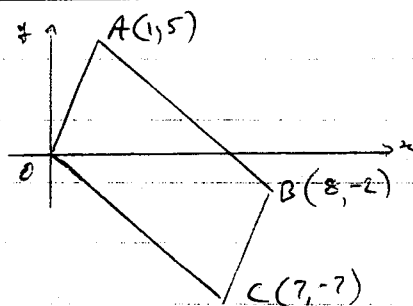
d) primitive of  $2 - x^{1/2} = 2x - \frac{2}{3} x^{3/2} + c$   
 $\textcircled{2}$

e)  $1 - 9x^2 = 0$   
 $(1 - 3x)(1 + 3x) = 0$   
 $x = \pm \frac{1}{3} \textcircled{2}$

f)  $112\frac{1}{2}\% = 115$   
 $1\% = 1.0222 \dots$   
 $100\% = 102.22 \dots$

Recommended retail  
 price is 102.2 cents/litre.  
 $\textcircled{2}$

QUESTION 2 (SH)



a) Midpoint AC = (4, -1)  $\textcircled{1}$

b) Slope of AB = -1  $\textcircled{1}$

c) Equ. AB,  
 $y - 5 = -1(x - 1)$   
 $y - 5 = -x + 1$   
 $x + y = 6 \textcircled{1}$

d) length AB =  $\sqrt{(8-1)^2 + (-2-5)^2} \textcircled{1}$   
 $= \sqrt{98}$   
 $= 7\sqrt{2} \text{ units } \textcircled{1}$

e) Slope of OC =  $\frac{-7-0}{7-0}$   
 $= -1 \textcircled{1}$   
 $\therefore AB \parallel OC$

f) Slope OA = 5, Slope BC = 5  $\textcircled{1}$   
 Since OA  $\parallel$  BC and OC  $\parallel$  AB (from above)  
 then opposite sides are parallel  $\textcircled{1}$   
 and OABC is a parallelogram.

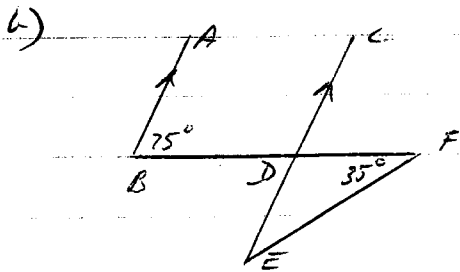
g)  $\perp d = \frac{|1 \times 0 + 1 \times 0 + 6|}{\sqrt{1^2 + 1^2}} \textcircled{1} = \frac{6}{\sqrt{2}} = 3\sqrt{2}$   
 units  $\textcircled{1}$

h) Area OABC =  $7\sqrt{2} \times 3\sqrt{2} \textcircled{1}$   
 $= 42 \text{ units}^2 \textcircled{1}$

### QUESTION 3 (AM)

a) i)  $\frac{d}{dx} x^2 e^{3x} = e^{3x} \times 2x + x^2 \times 3e^{3x}$   
 $= 2xe^{3x} + 3x^2 e^{3x}$  (2)  
 $= xe^{3x}(2 + 3x)$

ii)  $\frac{d}{dx} \frac{\sin x}{x} = \frac{x \times \cos x - \sin x \times 1}{x^2}$   
 $= \frac{x \cos x - \sin x}{x^2}$  (2)



$\angle CDF = 75^\circ$  Corresponding angles.  
 $\angle DEF + 35^\circ = 70^\circ$  Exterior angle  $\triangle DEF$   
 $\therefore \angle DEF = 40^\circ$  (2)

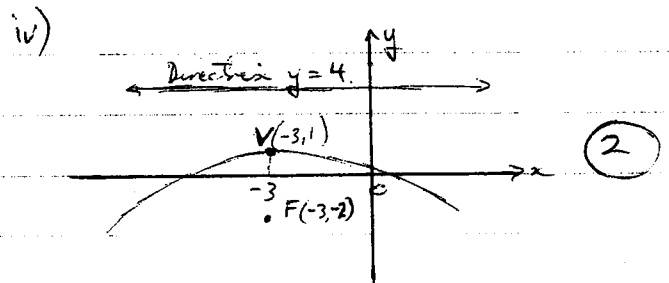
c)  $f'(x) = 8x^2 + 4$   
 $f(x) = 3x^3 + 4x + C$   
 $f(x) = 12$  when  $x = 2$   
 $\therefore 12 = 24 + 8 + C \Rightarrow C = -20$   
 $\therefore f(x) = 3x^3 + 4x - 20$  (2)

d)  $\int \frac{4x}{x^2 - 3} dx = 2 \log_e(x^2 - 3) + C$  (1)

e)  $\int_{\pi/6}^{\pi} \cos 2x dx = \left[ \frac{1}{2} \sin 2x \right]_{\pi/6}^{\pi}$  (1)  
 $= \frac{1}{2} \left( 0 - \frac{\sqrt{3}}{2} \right)$   
 $= -\frac{\sqrt{3}}{4}$  (1)

### QUESTION 4 (IB)

a)  $(x+3)^2 = -12(y-1)$   
 i) Vertex =  $(-3, 1)$  (1)  
 ii) Focus =  $(-3, -2)$  (1)  
 iii) Directrix is  $y = 4$  (1)

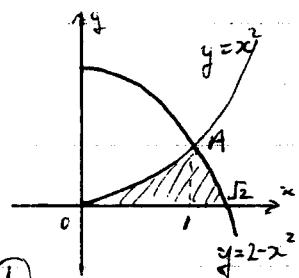


b) i)  $0.\dot{2}\dot{5} = \frac{25}{100} + \frac{25}{10000} + \frac{25}{1000000} + \dots$  (1)

ii) Using  $S_{\infty} = \frac{a}{1-r}$ ,  
 $0.\dot{2}\dot{5} = \frac{25}{100} \times \frac{100}{99}$   
 $= \frac{25}{99}$  (1)

c) i)  $x^2 = 2 - x^2$   
 $2x^2 - 2 = 0$   
 $x^2 - 1 = 0$   
 $x = \pm 1$

$\therefore$  x-coordinate of A is 1. (1)



ii) Area =  $\int_0^1 x^2 dx + \int_1^{\sqrt{2}} 2 - x^2 dx$  (1)

$= \left[ \frac{x^3}{3} \right]_0^1 + \left[ 2x - \frac{x^3}{3} \right]_1^{\sqrt{2}}$  (1)

$= \left( \frac{1}{3} - 0 \right) + \left[ \left( 2\sqrt{2} - \frac{2\sqrt{2}}{3} \right) - \left( 2 - \frac{1}{3} \right) \right]$

$= \frac{4\sqrt{2}}{3} - \frac{4}{3}$  (1)

$= 0.55$  (2 d.p.) (1)

## QUESTION 5 (1M)

a) i)  $f(x) = 2x^3 - 6x^2 - 18x + 1$

$f'(x) = 6x^2 - 12x - 18$

St. pts when  $6x^2 - 12x - 18 = 0$

$x^2 - 2x - 3 = 0$

$(x-3)(x+1) = 0$

$x = 3, -1$  (1)

Stationary pts  $(3, -53)$  &  $(-1, 11)$  (1)

$f''(x) = 12x - 12$

at  $x = 3$ ,  $f''(3) = 24 > 0$  (1)

$\therefore$  Minimum at  $(3, -53)$

at  $x = -1$ ,  $f''(-1) = -24 < 0$  (1)

$\therefore$  Maximum at  $(-1, 11)$

ii) Increasing  $f'(x) > 0$

$\therefore 6x^2 - 12x - 18 > 0$

$(x-3)(x+1) > 0$

$\therefore x < -1$  or  $x > 3$  (2)

iii) Rate of decrease greatest

when  $f''(x) = 0$   $\therefore x = 1$  (1)

c) 10, 15, 20, ...

i)  $T_n = a + (n-1)d$

$T_{10} = 10 + 9 \times 5$  (2)

$T_{10} = 55$   $\therefore$  Width 10<sup>th</sup> strip is 55cm

ii) Width of all strips =  $455 - 5$

$= 450$  (1)

$\therefore 450 = \frac{n}{2} [20 + (n-1)5]$

$900 = 15n + 5n^2$

$\therefore n^2 + 3n - 180 = 0$  (1)

$(n+15)(n-12) = 0$

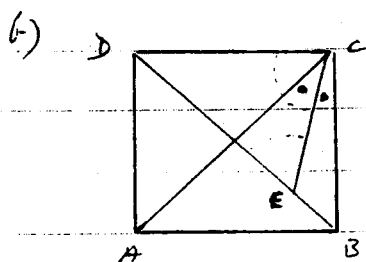
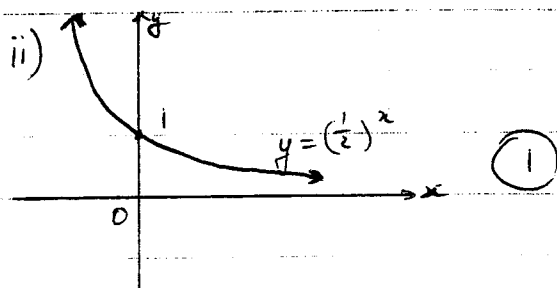
$\therefore$  12 pieces sewn together. (1)

## QUESTION 6 (DS)

a)  $\log_e y = x \log_e \left(\frac{1}{2}\right)$

i)  $\log_e y = \log_e \left(\frac{1}{2}\right)^x$

$\therefore y = \left(\frac{1}{2}\right)^x$  or  $y = 2^{-x}$  (2)



i) To prove  $\angle DCE = \angle DEC$

$\angle DCA = 45^\circ$  (Diagonals of a square bisect angles.)

Similarly  $\angle DBC = 45^\circ$

$\therefore \angle DEC = 45^\circ + \theta$  (Exterior angle of  $\triangle CBE$ .)

$\angle DCE = 45^\circ + \theta$  (adjacent angles.)

$\therefore \angle DEC = \angle DCE$  (2)

ii)  $\triangle DEC$  is isosceles (equal base angles.)

$\therefore DC = DE$  (sides opposite equal angles are equal)

But  $DC = DA$  (sides of square)

$\therefore DE = DA$  (2)

c) i)  $V = Ae^{-kt}$

$\frac{dV}{dt} = -kAe^{-kt}$  (1)

$\therefore \frac{dV}{dt} = -kV$  ( $V = Ae^{-kt}$ )

## QUESTION 6 CONT'D.

c) ii)  $v = Ae^{-kt}$

$t=0, v=85$

$\therefore 85 = Ae^0$

$\therefore A = 85$  (1)

$v = 85e^{-kt}$

$t=5, v=60$

$\therefore 60 = 85e^{-5k}$

$\frac{60}{85} = e^{-5k}$

$\ln\left(\frac{60}{85}\right) = -5k$

$k \doteq 0.0696613 \dots$  (2)

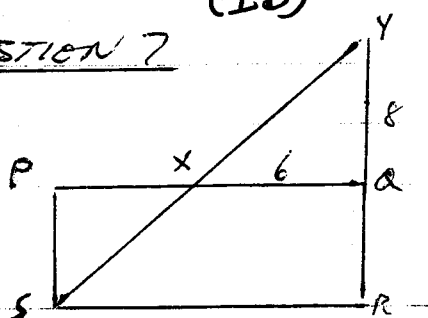
iii) When  $t=8$

$v = 85e^{-8k}$

$v \doteq 48.68 \text{ cm/s}$  (1)

## QUESTION 7

a)



In  $\triangle PXS$  &  $\triangle QXY$

i)  $\angle PXS = \angle QXY$  (Vertically opposite)

$\angle SPX = \angle XQR$  (right angles of rect. PQRS)

$= \angle YQX$  (straight angle)

$\therefore \triangle PXS \parallel \triangle QXY$  (2 equal angles)

(2)

ii)  $\frac{PS}{YA} = \frac{PX}{QX}$  corresponding sides of similar  $\triangle$ s.

$\therefore \frac{x}{8} = \frac{3x-6}{6}, PS=x$

$6x = 24x - 48$  (2) Note:  $SR = 3PS$

$18x = 48 \therefore x = \frac{8}{3}, PS = \frac{8}{3}$

b)  $v = 1 - 2\cos t \quad 0 \leq t \leq 2\pi$

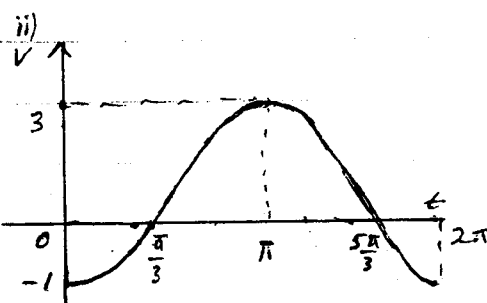
i) At rest,  $v=0$ .

$1 - 2\cos t = 0$

$\cos t = \frac{1}{2}$

$t = \frac{\pi}{3}, \frac{5\pi}{3}, \dots$  (2)

At rest when  $t = \frac{\pi}{3}$  and  $\frac{5\pi}{3}$  secs.



(2)

iii) Maximum velocity is  $3 \text{ m/s}$  (1)

iv) Distance travelled is given by

$\int_0^{\pi/3} -(1-2\cos t) dt + \int_{\pi/3}^{\pi} (1-2\cos t) dt$  (1)

distance =  $\int_0^{\pi/3} 2\cos t - 1 dt + \int_{\pi/3}^{\pi} 1 - 2\cos t dt$

$= [2\sin t - t]_0^{\pi/3} + [t - 2\sin t]_{\pi/3}^{\pi}$  (1)

$= (2 \times \frac{\sqrt{3}}{2} - \frac{\pi}{3}) - (0) + (\pi - 0) - (\frac{\pi}{3} - 2 \times \frac{\sqrt{3}}{2})$

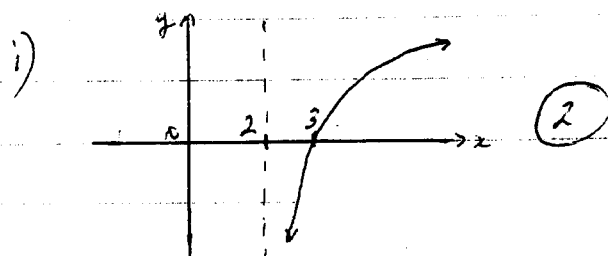
$= \sqrt{3} - \frac{\pi}{3} + \pi - \frac{\pi}{3} + \sqrt{3}$

$= 2\sqrt{3} + \frac{\pi}{3} \text{ metres}$  (1)

# QUESTION 8

(IB)

a)  $y = \ln(x-2) \quad x > 2$



ii)

$$\int_3^5 \ln(x-2) dx \approx \frac{h}{3} [y_0 + y_2 + 4y_1] \text{ S.R. 3 f.u.}$$

$$= \frac{1}{3} [0 + \ln 3 + 4 \ln 2] \quad (1)$$

$$= 1.3 \text{ (1 d.p.)} \quad (1)$$

b)  $x^2 + mx + n = 0$

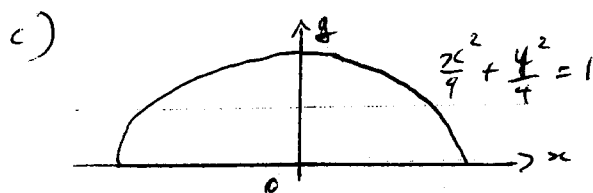
Let the roots be  $\alpha$  and  $2\alpha$

$\therefore$  Sum,  $3\alpha = -m \quad (1)$

Product,  $2\alpha^2 = n \quad (1)$

$$\therefore \frac{m^2}{n} = \frac{9\alpha^2}{2\alpha^2} \quad (1)$$

$$= \frac{9}{2} \quad (1)$$



i) when  $y=0$ ,  $\frac{x^2}{9} + 0 = 1$

$$x^2 - 9 = 0$$

$$x = \pm 3 \quad (1)$$

ii)  $V = \pi \int_a^b y^2 dx$

Note:  $y^2 = 4 - \frac{4x^2}{9}$

$$\therefore V = 2\pi \int_0^3 \left(4 - \frac{4x^2}{9}\right) dx \quad (1)$$

$$= 2\pi \left[4x - \frac{4}{27}x^3\right]_0^3 \quad (1)$$

$$= 2\pi [(12 - 4) - 0]$$

$$\therefore \text{Volume} = 16\pi \text{ units}^3 \quad (1)$$

# QUESTION 9 (SH)

a)



i)  $P(\text{sum is zero}) = \frac{1}{6} \quad (1)$

ii)  $P(\text{one card is 2}) = \frac{1}{2} \quad (1)$

b)  $L = 1000(20-t)^2$

i) when  $t=0$ ,  $L = 400000$

$\therefore 400000$  L of water initially.  $(1)$

ii)  $L = 1000(20-t)^2$

$$\frac{dL}{dt} = 2 \times 1000(20-t)^1 \times -1$$

$$= -2000(20-t)$$

$$= -40000 + 2000t$$

$$= 2000t - 40000 \quad (1)$$

# QUESTION 9 cont'd.

$$\text{iii)} -20000 = 2000t - 40000$$

$$20000 = 2000t$$

$$t = 10$$

Water is flowing out at 20 000 L/min  
when  $t=10$  min. (1)

iv) 20 min to empty tank (1)

v) Tank half full,

$$200000 = 1000(20-t)^2$$

$$200 = (20-t)^2$$

$$\pm 10\sqrt{2} = 20-t$$

$$\therefore t = 20 - 10\sqrt{2} \text{ or } (20 + 10\sqrt{2}) \quad 0 \leq t \leq 20$$

$$\therefore t = 20 - 10\sqrt{2} \text{ or } 5.86 \text{ min 2dp}$$

$$\therefore \frac{dL}{dt} = 2000(20 - 10\sqrt{2}) - 40000$$

$$(2) = -20000\sqrt{2} \text{ or } -28284.27$$

$\therefore$  tank is emptying at a rate  
of 20 000 $\sqrt{2}$  L/min when half full.

$$c) f(x) = \pi x - \cos \pi x$$

$$i) f'(x) = \pi + \pi \sin \pi x$$

$$f''(x) = \pi^2 \cos \pi x$$

$$f''\left(\frac{1}{2}\right) = 0$$

(2)

$$ii) \text{ when } x = \frac{1}{2}, f(x) = \frac{\pi}{2} - \cos \frac{\pi}{2}$$

$$= \frac{\pi}{2}$$

$$\therefore \left(\frac{1}{2}, \frac{\pi}{2}\right) \text{ lies on } f(x) \quad (1)$$

$$f''\left(\frac{1}{2}\right) = 0$$

$$f''\left(\frac{1}{2}-\epsilon\right) > 0 \text{ i.e. } \pi^2 \cos \frac{\pi}{4} > 0$$

$$f''\left(\frac{1}{2}+\epsilon\right) < 0 \text{ i.e. } \pi^2 \cos \frac{3\pi}{4} < 0$$

Since concavity changes about  $x = \frac{1}{2}$   
then  $\left(\frac{1}{2}, \frac{\pi}{2}\right)$  is a pt of inflection.

(1)

## QUESTION 10.

(IM).

$$a) i) A = P\left(1 + \frac{r}{100}\right)^n$$

$$= 2000\left(1 + \frac{4}{100}\right)^{40}$$

$$= 2000(1.04)^{40}$$

$$= \$9602.04$$

(1)

(1)

ii)

$$\text{Amount} = 500 \times 1.04^{38} + 500 \times 1.04^{36} + \dots + 500 \times 1.04^2$$

(1)

$$= \frac{500 \times 1.04^2 [(1.04^2)^{19} - 1]}{(1.04^2)^2 - 1}$$

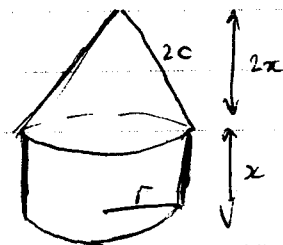
(1)

$$= \$22790.57$$

(1)

$\therefore$  Amount in account at end of  
20 years is  $\$9602.04 + \$22790.57$   
i.e.  $\$32392.61$  (1)

4)



$$i) r^2 + (20-x)^2 = 20^2$$

$$r^2 + 4x^2 = 400$$

$$\therefore r^2 = 400 - 4x^2$$

(1)

QUESTION 10 CONT'D.

$$\begin{aligned}\text{ii) Volume of silo, } V &= \pi r^2 h + \frac{1}{3} \pi r^2 h \\ &= \pi r^2 x + \frac{1}{3} \pi r^2 (2x) \\ &= \frac{5}{3} \pi r^2 x \\ &= \frac{5}{3} \pi x (400 - 4x^2) \\ &= \frac{20}{3} \pi (100x - x^3) \quad (1)\end{aligned}$$

$$\begin{aligned}\text{iii) } \frac{dV}{dx} &= \frac{20}{3} \pi (100 - 3x^2) \\ \frac{d^2V}{dx^2} &= \frac{20}{3} \pi (-6x) \quad (1) \\ &= -40 \pi x\end{aligned}$$

Stationary pts,  $\frac{dV}{dx} = 0$

$$\begin{aligned}\text{ie. } \frac{20\pi}{3} (100 - 3x^2) &= 0 \\ 3x^2 &= 100 \\ x^2 &= \frac{100}{3} \\ x &= \frac{10}{\sqrt{3}} \quad (x > 0) \quad (1)\end{aligned}$$

$$\frac{d^2V}{dx^2} < 0 \text{ for } x = \frac{10}{\sqrt{3}} \quad (1)$$

$\therefore$  Maximum occurs when  $x = \frac{10}{\sqrt{3}}$

$$\begin{aligned}\therefore \text{required height} &= 3 \times \frac{10}{\sqrt{3}} \\ &= 10\sqrt{3} \text{ metres.} \\ &\quad (1)\end{aligned}$$